

CUSTOMER SEGMENTATION DASHBOARD USING MACHINE LEARNING

Abstract

Customer segmentation is a critical business strategy that enables organizations to understand customer behavior, identify valuable customer groups, and develop targeted marketing campaigns. This project presents a Customer Segmentation Dashboard built using Machine Learning techniques to classify customers into distinct groups based on demographic, financial, and behavioral characteristics. The project includes data preprocessing, exploratory data analysis, dimensionality reduction using Principal Component Analysis (PCA), clustering using K-Means, Hierarchical Clustering, and DBSCAN, customer persona development, marketing strategy recommendations, and an interactive Streamlit dashboard. The solution helps businesses make data-driven decisions, improve customer retention, increase revenue, and enhance customer engagement.

Chapter 1: Introduction

1.1 Background

Organizations collect large amounts of customer data from various sources. However, extracting actionable insights from this data is often challenging. Customer segmentation helps businesses group customers with similar characteristics, enabling personalized marketing strategies and better resource allocation.

Machine Learning clustering techniques provide an effective way to automatically identify customer groups without requiring predefined labels.

1.2 Problem Statement

Businesses often struggle to understand customer diversity and purchasing behavior. Without proper segmentation, marketing efforts become less effective and customer engagement decreases.

The primary objective of this project is to develop a machine learning-based customer segmentation system capable of:

- Identifying distinct customer groups
- Understanding customer behavior patterns
- Creating customer personas
- Supporting targeted marketing campaigns
- Improving customer retention and profitability

1.3 Objectives

The main objectives of this project are:

- Perform customer data preprocessing
 - Conduct exploratory data analysis
 - Apply feature engineering techniques
 - Implement clustering algorithms
 - Identify optimal customer segments
 - Develop customer personas
 - Generate marketing recommendations
 - Build an interactive dashboard for visualization
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Chapter 2: Dataset Description

2.1 Dataset Source

Dataset: Customer Segmentation Dataset

Source: Kaggle

2.2 Dataset Overview

The dataset contains customer demographic, financial, and behavioral information.

Total Records: 53,503

Features Include:

- Customer ID
- Age
- Gender
- Marital Status
- Education Level
- Geographic Information
- Occupation
- Income Level
- Coverage Amount
- Premium Amount

- Policy Type
- Customer Preferences
- Communication Channels
- Segmentation Group

2.3 Business Importance

These features provide valuable insights into customer purchasing patterns, insurance preferences, and communication behaviors, making them suitable for customer segmentation analysis.

Chapter 3: Data Preprocessing

3.1 Data Inspection

The dataset was loaded using Pandas and inspected for structure, data types, and summary statistics.

Activities performed:

- Shape Analysis
- Data Type Verification
- Null Value Detection
- Duplicate Detection

3.2 Missing Value Analysis

Missing values were analyzed using Pandas functions.

Result:

- No significant missing values found.

3.3 Duplicate Detection

Duplicate records were checked and removed if present.

Result:

- Dataset contained no critical duplicate entries.

3.4 Feature Engineering

Additional business-oriented features were created:

Age Group

Customers categorized into:

- Young
- Adult
- Middle Age
- Senior

Income Level Category

Income divided into:

- Low Income
- Medium Income
- High Income

Spending Category

Customers categorized based on premium spending patterns.

3.5 Data Encoding

Categorical features were converted into numerical representations using Label Encoding.

3.6 Feature Scaling

StandardScaler was used to normalize feature values before clustering.

Benefits:

- Prevents dominance of large-scale variables
- Improves clustering performance
- Enhances model stability

Chapter 4: Exploratory Data Analysis

4.1 Dataset Statistics

Descriptive statistics were generated to understand data distribution.

Observations:

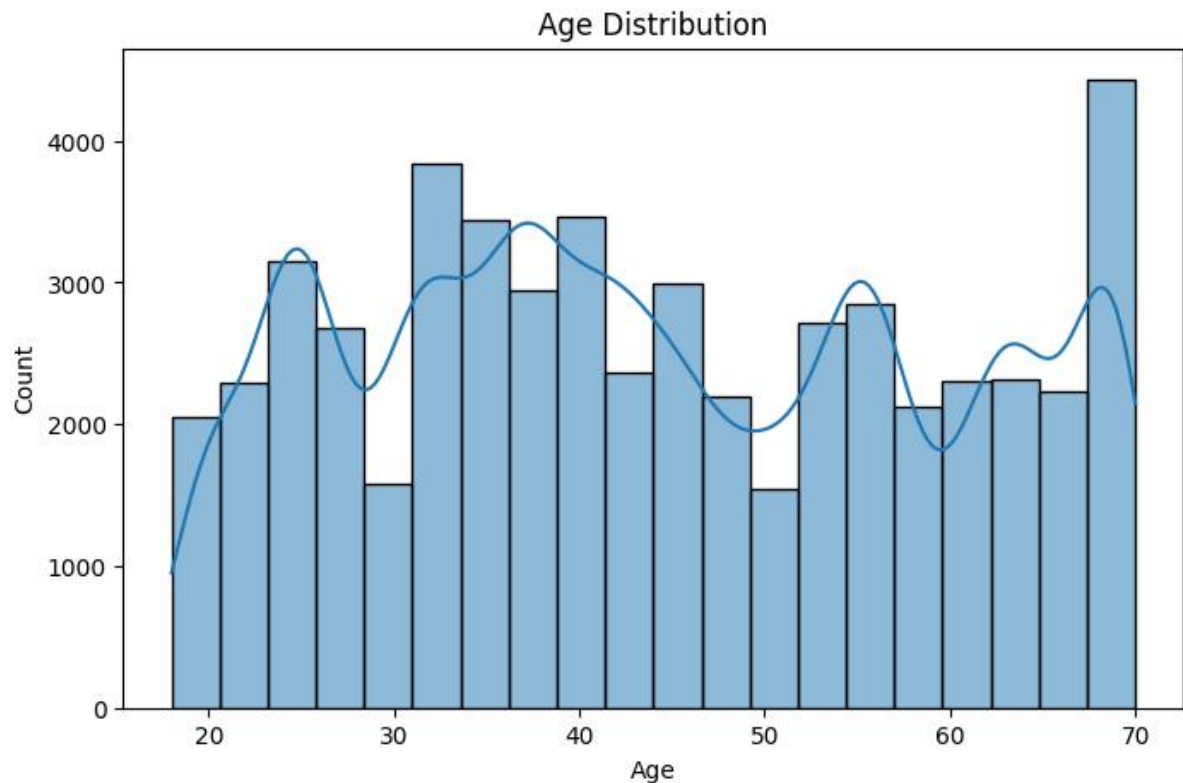
- Wide variation in customer income
- Significant variation in coverage amounts
- Diverse premium payment patterns

4.2 Age Distribution Analysis

The age distribution revealed that most customers belong to adult and middle-aged groups.

Business Insight:

These groups represent the largest market opportunity for targeted insurance products.



4.3 Income Distribution Analysis

Income levels varied significantly across customers.

Business Insight:

Different pricing strategies can be designed for different income groups.

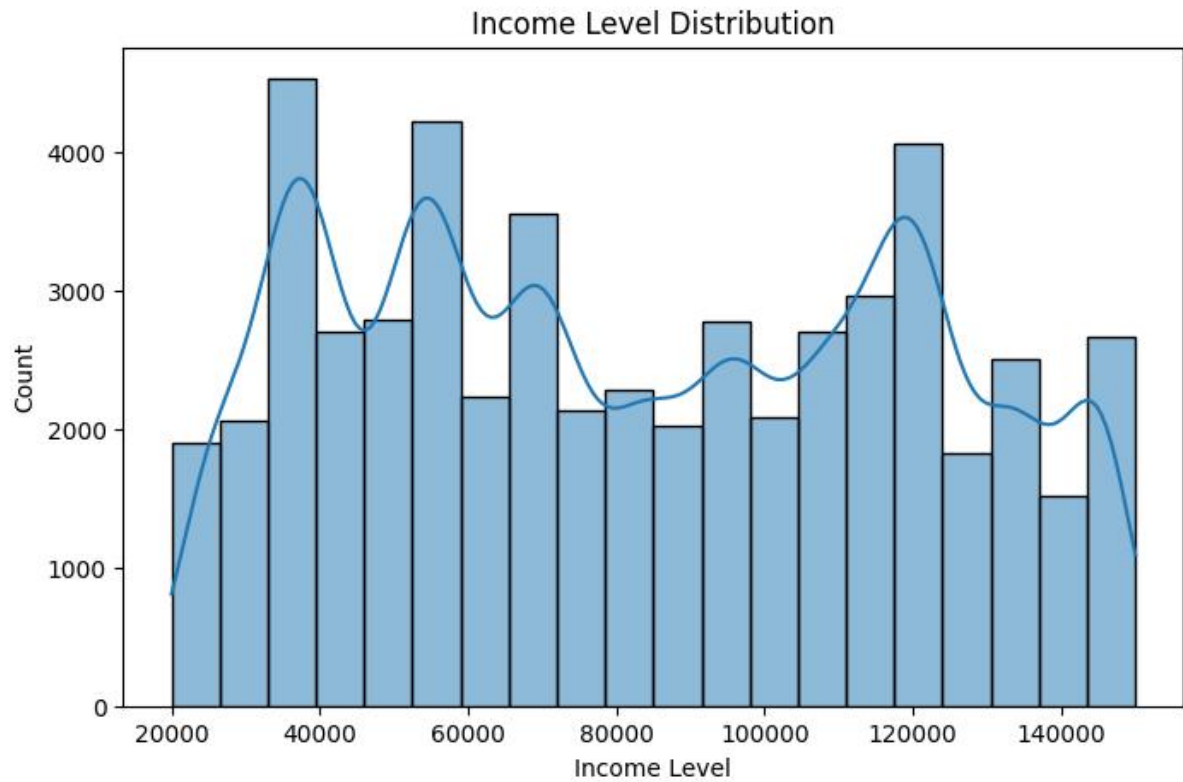
[Insert Income Distribution Screenshot]

4.4 Coverage vs Premium Analysis

A positive relationship was observed between coverage amount and premium amount.

Business Insight:

Customers purchasing higher coverage plans contribute significantly to revenue.

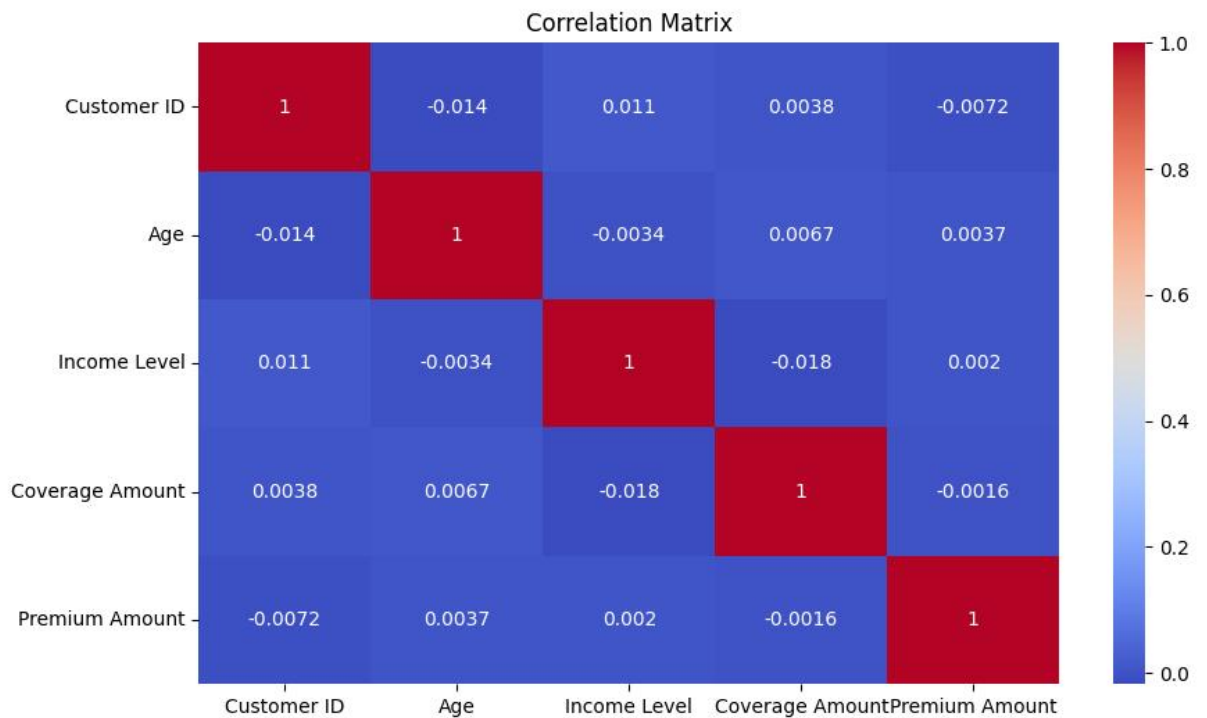


4.5 Correlation Analysis

A correlation heatmap was generated.

Key Findings:

- Coverage Amount positively correlates with Premium Amount.
- Income influences customer purchasing behavior.



Chapter 5: Principal Component Analysis (PCA)

5.1 Purpose of PCA

Principal Component Analysis was applied to:

- Reduce dimensionality
- Remove redundancy
- Improve visualization
- Enhance clustering performance

5.2 PCA Results

The first two principal components captured the majority of data variation.

Benefits:

- Simplified visualization
- Faster model execution
- Improved cluster interpretation

Chapter 6: Customer Segmentation

6.1 Clustering Algorithms

Three clustering algorithms were implemented:

K-Means Clustering

Advantages:

- Fast
- Scalable
- Easy to interpret

Hierarchical Clustering

Advantages:

- Visual hierarchy
- Flexible cluster structure

DBSCAN

Advantages:

- Detects noise
- Handles irregular cluster shapes

6.2 Optimal Cluster Selection

The Elbow Method and Silhouette Score were used to determine the optimal number of clusters.

Result:

Five customer segments provided the best clustering performance.

6.3 K-Means Results

K-Means was selected as the final model because it produced well-separated and interpretable customer segments.

Chapter 7: Customer Personas

Based on clustering results, five customer personas were developed.

Persona 1: Premium Protection Customers

Characteristics:

- High Income

- High Coverage
- Premium Products

Marketing Strategy:

- VIP Membership
- Personalized Service
- Exclusive Offers

Persona 2: Value Seekers

Characteristics:

- Moderate Income
- Price Sensitive

Marketing Strategy:

- Discounts
- Bundled Packages

Persona 3: Wealthy Low-Risk Customers

Characteristics:

- Very High Income
- Stable Customers

Marketing Strategy:

- Premium Insurance Plans
- Wealth Protection Services

Persona 4: Affluent Growth Customers

Characteristics:

- High Growth Potential

Marketing Strategy:

- Upselling Campaigns
- Personalized Recommendations

Persona 5: Senior Loyal Customers

Characteristics:

- Long-Term Customers
- Strong Retention Potential

Marketing Strategy:

- Loyalty Programs
 - Renewal Benefits
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Chapter 8: Marketing Recommendations

8.1 Retention Strategies

- Loyalty Rewards
- Personalized Communication
- Dedicated Customer Support

8.2 Upselling Strategies

- Premium Policy Upgrades
- Higher Coverage Plans
- Exclusive Membership Programs

8.3 Cross-Selling Strategies

- Additional Insurance Products
- Family Insurance Packages
- Investment-Linked Products

8.4 Email Campaigns

Recommended Campaigns:

- Welcome Series
- Renewal Reminders
- Loyalty Promotions
- Premium Upgrade Campaigns

8.5 Social Media Campaigns

Platforms:

- Facebook

- Instagram
- LinkedIn
- YouTube
- WhatsApp

Benefits:

- Increased engagement
 - Higher conversion rates
 - Improved customer retention
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Chapter 9: Dashboard Development

Technologies Used

- Streamlit
- Plotly
- Pandas
- Scikit-Learn

Dashboard Features

Executive Dashboard

Displays:

- Customer Count
- Average Income
- Cluster Statistics

EDA Dashboard

Displays:

- Histograms
- Boxplots
- Correlation Heatmaps

Segmentation Dashboard

Displays:

- PCA Visualization
- Cluster Distribution
- Cluster Profiles

Personas Dashboard

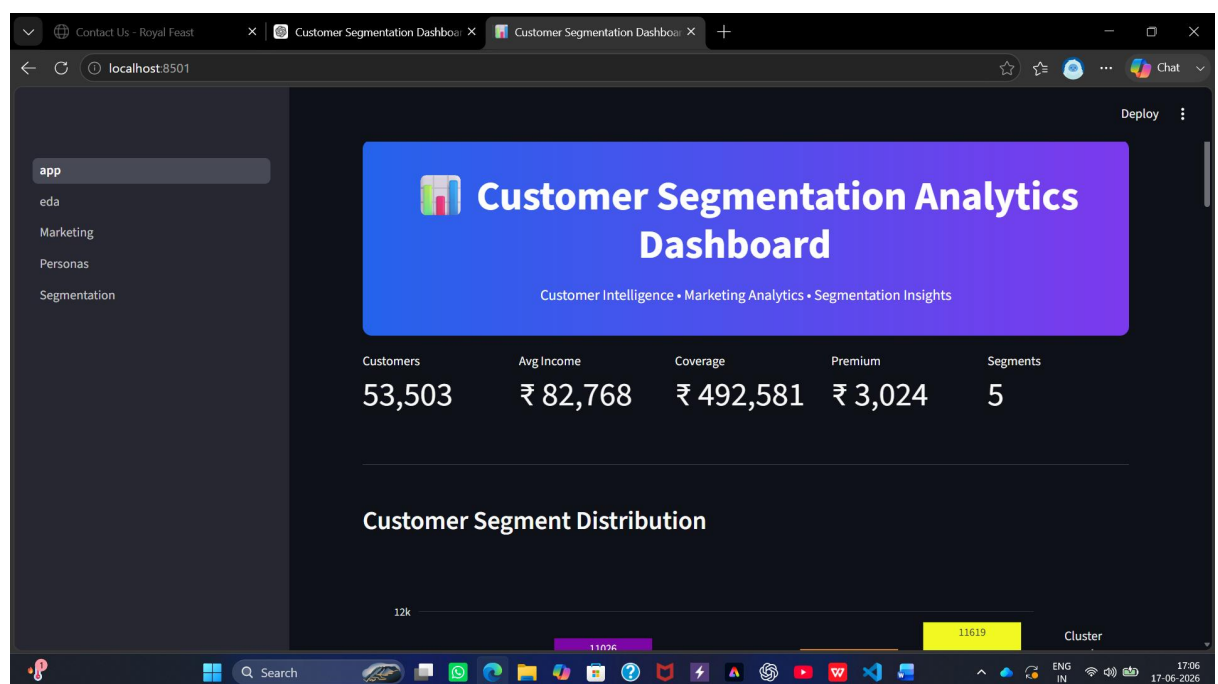
Displays:

- Customer Personas
- Behavioral Characteristics

Marketing Dashboard

Displays:

- Marketing Recommendations
- Retention Strategies
- Campaign Suggestions



Chapter 10: Results and Conclusion

Results

The project successfully:

- Processed customer data

- Identified customer segments
- Generated customer personas
- Produced marketing recommendations
- Built an interactive analytics dashboard

Business Benefits

- Improved customer understanding
- Better targeting strategies
- Increased retention opportunities
- Enhanced revenue generation
- Data-driven marketing decisions

Future Enhancements

- Real-Time Customer Segmentation
- Customer Lifetime Value Prediction
- Recommendation Engine
- AI-Powered Marketing Automation
- Cloud Deployment

Conclusion

The Customer Segmentation Dashboard successfully demonstrates how machine learning can be applied to customer analytics. By combining clustering algorithms, customer personas, marketing insights, and interactive visualizations, the system provides valuable business intelligence that can support strategic decision-making and improve customer relationship management.

References

1. Scikit-Learn Documentation
2. Pandas Documentation
3. Streamlit Documentation
4. Plotly Documentation
5. Kaggle Customer Segmentation Dataset
6. Python Official Documentation